

AgriAgent

Alternatives Dossier & 2026 Model Landscape

Deep research report on 5 alternative FYP ideas and the current model landscape.
Each alternative follows the same dossier structure: problem, pipeline, datasets, papers, metrics, feasibility, publishability, risks, and roadmap.

Instructor	Prof. V. D. Dhore
Framework	SmartDesk (LangGraph) + Grounding DINO + SAM 2
Alternatives Covered	Drone Scout, Smart Harvester, Sticky Trap, Post-Harvest, Hydroponics
2026 Models Discussed	SAM 3, Grounding DINO 1.5/2, Qwen2.5-VL, Florence-2

A. 2026 Model Landscape & Justification

This section addresses the critical question that Prof. Dhore and thesis reviewers will raise: "Why use models from 2024 when it is 2026?" The answer is nuanced and requires honest assessment. Both Grounding DINO and SAM 2 are already vision-transformer models (Swin Transformer and Hiera ViT, respectively), not legacy CNNs. However, newer alternatives have emerged that must be discussed.

A.1 SAM 3: Segment Anything with Concepts (November 2025)

Release: November 2025 by Meta AI. SAM 3 is a unified foundation model for promptable segmentation that accepts **text prompts** (open-vocabulary short noun phrases like "segment the weed") and image exemplar prompts, eliminating the need for bounding box inputs. SAM 3.1 was announced March 2026 with faster real-time video segmentation. The model achieves 825 citations on OpenReview within months. **GitHub:** <https://github.com/facebookresearch/sam3>. **Ultralytics integration:** Native support since October 2025.

Critical Implication for AgriAgent: SAM 3 could theoretically collapse the two-stage pipeline (Grounding DINO produces boxes, SAM 2 converts boxes to masks) into a single model that segments directly from text. This is the strongest counter-argument a reviewer could raise. **However**, SAM 3 is significantly heavier than SAM 2-Tiny (38.9M params) and will likely OOM on a free Google Colab T4 GPU (16GB VRAM). The smallest SAM 3 variant requires approximately 4-6GB VRAM for inference alone, leaving insufficient headroom for Grounding DINO and the LangGraph orchestrator simultaneously. For an FYP constrained to free hardware, SAM 2-Tiny remains the pragmatic choice.

A.2 Grounding DINO Evolution (2024-2026)

The original Grounding DINO (ECCV 2024) remains the standard open-vocabulary detector. There is no formal "Grounding DINO 2" release, but the IDEA-Research team has released **Grounding DINO 1.5 Edge** with improved edge deployment and **Dynamic-DINO** (ICCV 2025, cited by 9) which extends Grounding DINO 1.5 with a Mixture-of-Experts (MoE) tuning strategy for real-time open-vocabulary detection. The original Grounding DINO-T (172M params) remains the most documented and HuggingFace-supported variant, with native Transformers integration since April 2024.

Recommendation: Use Grounding DINO-T as the primary detector. Attempt Grounding DINO 1.5 Edge as an upgrade in Week 2; if it gives compatibility issues on Colab T4, fall back to Grounding DINO-T. Both are Swin-Transformer-based open-vocabulary detectors. Document the comparison in the paper.

A.3 VLM-Guided Refinement (The 2026 Flourish)

Qwen2.5-VL (January 2025, Alibaba) is a flagship multimodal model that processes images at native resolution with bounding box and point-based grounding. Available in 3B, 7B, and 72B parameter sizes. **Qwen3-VL-Seg** (May 2026) converts open-world grounding priors into pixel-level segmentation. **Florence-2** (Microsoft) provides unified vision tasks including grounding. These VLMs represent the 2026 trend of models that can both detect and reason about visual content.

Novel 2026 Contribution: Instead of using a fixed heuristic (if IoU < 0.85, add corrective point), integrate a lightweight VLM (Qwen2.5-VL-3B via Groq API or local inference) as a **"reasoning critic"** in the refinement loop. The VLM looks at the mask + image and decides whether the segmentation is correct, suggesting where to correct based on visual reasoning rather than a fixed threshold. This is genuinely novel: "VLM-guided refinement instead of RL" is directly comparable to MedSAM-Agent but simpler and more interpretable. The 3B variant fits in CPU memory and can run alongside the GPU pipeline.

A.4 Strategic Options

Option	Risk	Pros	How to Defend
A: Keep SAM 2 + GDINO-T	LOW	Runs on free T4, mature, bulletproof demo	Discuss SAM 3/VLMs in Limitations section
B: Upgrade to GDINO 1.5 + SAM 2.1	LOW-MED	"Latest stable" stack, better zero-shot	Swap checkpoints; keep architecture identical
C: SAM 3 / unified VLM	HIGH	Cutting-edge but OOM risk on T4	Only if paid A100 GPU access available

Table A1: Strategic options for model selection.

Recommended: Option B as a light upgrade, defended like Option A. Keep SAM 2-Tiny (T4-safe). Try Grounding DINO 1.5 Edge; fall back if needed. Add VLM-guided refinement as the 2026 novelty. In the paper, dedicate a subsection to "Why not SAM 3 / unified VLMs?" answering with compute constraints and precision trade-offs. Pre-empting this question separates an accepted paper from a rejected one.

A.5 Defense Script for Prof. Dhore

"Both Grounding DINO and SAM 2 are vision-transformer foundation models, not legacy CNNs. We chose them because they run on free Colab T4 hardware, and our contribution is the agentic orchestration framework, not the underlying models. We explicitly compare and discuss newer 2025-26 models like SAM 3 and Grounding DINO 1.5 in our limitations section, and demonstrate VLM-guided refinement as a 2026 extension of our pipeline."

1. Alternative: Drone Scout (Macro-Level Weed Mapping)

1.1 Problem & Motivation

Farmers fly drones over fields but spend hours manually reviewing orthomosaic (stitched drone) images to locate weed patches. Manual annotation of drone imagery at field scale is impractical due to the sheer area covered, often hundreds of acres per flight. Existing supervised approaches like WeedNet (Castellano et al., 2023, cited by 58) use lightweight Vision Transformers on multispectral drone imagery but require retraining for each new field. The WeedMap dataset (Sa et al., 2018, cited by 416) from ETH Zurich covers 16,550 m² of sugar beet fields with multispectral orthomosaic images from RedEdge-M and Sequoia cameras. A 2026 paper on Active Informative Planning for UAV-based Weed Mapping (arXiv:2601.13196) still uses WeedMap as the standard benchmark, confirming its relevance. Moving from micro (leaf-level) to macro (field-level) segmentation is a massive trend in precision agriculture, making this alternative highly publishable.

1.2 Pipeline

Input: Orthomosaic image (stitched from drone photos, typically 10000x10000+ pixels) + text prompt. **Grounding Agent:** Receives "weed patch. crop row. bare soil." and produces bounding boxes. Grounding DINO handles large images via sliding window or tiling (the orthomosaic must be split into tiles that fit the model input size, typically 1024x1024). **Segmentation Agent:** SAM 2 converts each box into a pixel mask per tile. **Analysis Agent:** Computes GPS coordinates of weed patches from image geotags + pixel coordinates, calculates total weed area per patch, and generates a flight path for a spray drone to visit only those coordinates. **Report Agent:** Outputs a georeferenced spray map (GeoJSON) + estimated flight time + chemical savings.

1.3 Datasets

Dataset	Content	Citations	URL	Role
WeedMap (ETH Zurich)	Multispectral orthomosaic, 16,550 m ²	416 (Sa 2018)	https://github.com/viariasv/weedMap	Primary
SugarBeets 2016 (Bonn)	25,429 UGV images, pixel masks	453 (Chebrolu 2017)	ipb.uni-bonn.de/data/sugarbeets2016	Secondary

Dataset	Content	Citations	URL	Role
Drone Weed (CSIC)	Early-season weed from drone	8 (Mesias-Ruiz 2025)	digital.csic.es	Supplementary

Table 1.1: Datasets for Drone Scout alternative.

1.4 Key Papers

WeedMap (Sa et al., 2018, Remote Sensing, cited by 416): The primary benchmark. Multispectral orthomosaic weed mapping framework using FCN and SegNet. **Castellano et al. (2023, cited by 58):** Lightweight Vision Transformers for weed mapping in drone imagery. **Active Informative Planning (2026, arXiv:2601.13196):** UAV-based active weed mapping using the WeedMap dataset. **Weed Instance Segmentation from UAV ortho-mosaic (2024, bioRxiv):** Deep learning for weed identification in potato fields from UAV imagery. **Drone Weed Dataset (Mesias-Ruiz et al., 2025, cited by 8):** Early-season weed classification from drone imagery for maize and tomato.

1.5 Evaluation Metrics & Targets

Metric	Computation	Target
Weed patch detection	Precision / Recall / AP at IoU 0.5	>0.7 AP for weed patches
Segmentation quality	mIoU (weed vs crop vs soil)	>0.65 mIoU zero-shot
Spray path efficiency	Flight distance vs. full-field	>60% distance reduction
Chemical savings	$1 - (\text{weed_area} + \text{buffer}) / \text{total_area}$	>70% herbicide savings
Processing speed	Seconds per orthomosaic tile	<3 sec per 1024x1024 tile on T4

Table 1.2: Evaluation metrics and targets for Drone Scout.

1.6 Feasibility & Risks

GPU: Same as main project (Colab T4). Orthomosaic tiling adds complexity but no additional VRAM. **Data:** WeedMap is publicly available (GitHub). SugarBeets already in main plan. **Key Risk:** Orthomosaic images are very large (10K+ pixels); tiling logic must correctly stitch masks back together. Grounding DINO may struggle with the very different statistics of top-down drone imagery vs. close-up field photos. **Mitigation:** Test on WeedMap tiles in Week 2; if zero-shot performance is below 0.5 AP, add few-shot fine-tuning of Grounding DINO on 10-20 annotated tiles. **Timeline:** 14 weeks (narrower scope than 5-app main project). **Cost:** \$0.

1.7 Publishability

Primary Venue: MDPI Remote Sensing (IF ~5.0) or Computers and Electronics in Agriculture. **Novelty Hook:** Zero-shot orthomosaic weed mapping with multi-agent orchestration is novel. No existing work combines Grounding DINO + SAM 2 on drone imagery with LangGraph agent coordination. The macro-level application differentiates from the dominant micro-level (close-up) segmentation papers. **SDG Alignment:** SDG 15 (Life on Land) through reduced chemical runoff.

1.8 Condensed Roadmap

Weeks	Tasks	Lead
1-2	Setup + WeedMap download + tile orthomosaic into 1024x1024 patches	All
3-5	Grounding DINO + SAM 2 on WeedMap tiles; compute GPS coordinates	M1+M2
6-8	Spray path generation + flight optimization + refinement loop	M1+M3
9-11	Full evaluation on WeedMap + SugarBeets; statistical significance	M3

Weeks	Tasks	Lead
12-14	Streamlit demo (upload orthomosaic, view spray map) + thesis writing	M4+All

Table 1.3: Condensed 14-week roadmap.

2. Alternative: Smart Harvester (Fruit Ripeness Grading)

2.1 Problem & Motivation

Robotic fruit pickers need to distinguish ripe from unripe fruit and determine the exact shape for grasp planning without bruising. Current systems use fixed-threshold color analysis (e.g., HSV filtering for red tomatoes) that fails under variable lighting, occlusion by leaves, and mixed ripeness within a single cluster. AI-driven ripeness detection is an active 2026 research area: a University of Dayton thesis (April 2026) uses YOLOv8 for automated tomato ripeness detection; IEEE DataPort published a Multi-Class Strawberry Ripeness Detection Dataset in February 2026; and real-time strawberry ripeness classification was published in Agriculture (MDPI, 2025). The zero-shot angle is compelling: instead of training a classifier for each fruit type, the farmer types "ripe tomato" or "unripe strawberry" and the system adapts instantly.

2.2 Pipeline

Input: Orchard/greenhouse image + prompt. **Grounding Agent:** "ripe tomato. green tomato. ripe strawberry. unripe strawberry." produces bounding boxes with ripeness labels. **Segmentation Agent:** SAM 2 traces the exact contour of each fruit (critical for robotic grasp planning). **Analysis Agent:** Computes a "Ripeness Index" per row: $\text{count(ripe)} / \text{count(total)}$. Generates harvest schedule: "Row 4 is 90% ripe, send picking crew tomorrow; Row 7 is 40% ripe, wait 3 days." **Report Agent:** Outputs ripeness heatmap + harvest schedule CSV + per-fruit masks for robotic path planning.

2.3 Datasets

Dataset	Content	URL	Role
MinneApple (U of Minnesota)	1,000 images, 41K+ apple instances, instance seg masks	https://github.com/nicolaihaeni/MinneApple	Primary
Strawberry Ripeness (IEEE)	Multi-class ripeness detection, Feb 2026	ieee-dataport.org	Secondary
Tomato Ripeness (Kaggle)	Multiple ripeness stages, YOLO format	kaggle.com (search tomato ripeness)	Secondary
Roboflow Fruit Datasets	Various fruit detection datasets	universe.roboflow.com	Supplementary

Table 2.1: Datasets for Smart Harvester alternative.

2.4 Key Papers

MinneApple (Haeni et al., 2019): Benchmark for apple detection and segmentation. Over 41,000 annotated instances in 1,000 images. **Ebbing (2026, U of Dayton):** YOLOv8 for automated ripeness detection in tomato plants. **Liang et al. (2022, cited by 80):** Real-time grading of defect apples using BiSeNet V2 semantic segmentation. **Ryu et al. (2026, cited by 3):** Deep learning for Fuji apple defect identification (cracks, bruises, diseases, scars). **Fotouhi et al. (2024, cited by 10):** Persistent monitoring of insect-pests on sticky traps (related monitoring approach).

2.5 Evaluation Metrics & Targets

Metric	Computation	Target
Ripeness classification	Per-fruit accuracy (ripe vs unripe)	>85% accuracy zero-shot

Metric	Computation	Target
Instance segmentation	Mask AP at IoU 0.5 (COCO protocol)	>0.6 mask AP
Count accuracy	MAE = mean(pred_count - true_count)	MAE < 3 per image
Ripeness Index	Correlation with manual rating	r > 0.85
Grasp feasibility	% of masks with sufficient boundary quality	>90% smooth boundaries

Table 2.2: Evaluation metrics for Smart Harvester.

2.6 Feasibility & Risks

GPU: Colab T4 (same as main). **Data:** MinneApple freely available (GitHub + UMN Data Depository). **Key Risk:** Grounding DINO may confuse "ripe" and "unripe" fruit since the distinction is subtle (color gradient rather than category). **Mitigation:** Use more specific prompts like "red tomato" vs "green tomato" rather than "ripe" vs "unripe". Add color-space post-processing (HSV analysis on the mask region) as a secondary confidence check. **Timeline:** 14 weeks. **Cost:** \$0.

2.7 Publishability

Primary Venue: IEEE RA-L (Robotics and Automation Letters) or CASE (Conference on Automation Science and Engineering). **Novelty Hook:** Zero-shot ripeness segmentation with natural language control is novel. Existing fruit grading papers use fixed classifiers; the NL prompt approach ("find red tomatoes") enables instant adaptation to new fruit types without retraining. The graspability analysis (mask boundary quality) connects segmentation to robotics, which RA-L reviewers value highly.

2.8 Condensed Roadmap

Weeks	Tasks	Lead
1-2	Setup + MinneApple download + test Grounding DINO on apple images	All
3-5	Pipeline: "red apple. green apple." prompts + SAM 2 masks	M1+M2
6-8	Ripeness Index computation + harvest schedule generation	M1+M3
9-11	Evaluation on MinneApple + strawberry dataset; ripeness accuracy	M3
12-14	Demo (upload orchard photo, view ripeness map) + thesis	M4+All

Table 2.3: Condensed 14-week roadmap.

3. Alternative: Sticky Trap Auditor (Pest Counting)

3.1 Problem & Motivation

Farmers hang yellow sticky traps in greenhouses to monitor insect populations (whiteflies, thrips, aphids, fungus gnats). Manual counting is tedious, error-prone, and infrequent, leading to delayed pest outbreak detection. Li et al. (2021, cited by 205) constructed a detection model for whitefly and thrips from sticky trap images. Fotouhi et al. (2024, cited by 10) proposed persistent monitoring of insects on sticky traps through deep learning. Bockmann et al. (2021, cited by 49) demonstrated rapid, low-cost insect species differentiation from yellow sticky traps. The IP102 dataset (Wu et al., CVPR 2019, cited by 798) provides 75,000+ images across 102 insect pest categories, making it the largest publicly available pest recognition benchmark. Tiny object detection on noisy sticky trap backgrounds is notoriously difficult, making this alternative an excellent test of the refinement loop.

3.2 Pipeline

Input: Photo of yellow sticky trap + prompt. **Grounding Agent:** "whitefly. aphid. thrips. fruit fly. insect." produces bounding boxes. Tiny insects (<20 pixels) are the key challenge; Grounding DINO may need multi-scale inference (image pyramid at 1x, 2x, 4x). **Segmentation Agent:** SAM 2 segments each insect for size-based species differentiation (larger insects = likely aphids; tiny = likely thrips). **Analysis Agent:** Counts insects per trap, compares against a "Pest Threshold Database" stored in ChromaDB via SmartDesk's KnowledgeAgent RAG, and generates alerts: "Whitefly exceeded safety limit in Greenhouse B. Action: Apply neem oil within 48 hours." **Report Agent:** Outputs count table + trend chart + alert summary.

3.3 Datasets

Dataset	Content	URL	Role
IP102 (CVPR 2019)	75,222 images, 102 classes, 19K annotated	https://github.com/xpwu95/IP102	Primary
Sticky Trap Whitefly (Li 2021)	Whitefly + thrips from greenhouse traps	ScienceDirect (cited 205)	Secondary
Pest Monitoring (Fotouhi 2024)	Persistent insect monitoring on traps	Frontiers in Plant Science	Reference
Roboflow IP102	IP102 on Roboflow Universe with API access	universe.roboflow.com	Convenient

Table 3.1: Datasets for Sticky Trap Auditor.

3.4 Evaluation Metrics & Targets

Metric	Computation	Target
Insect counting	$MAE = \text{mean}(\text{pred_count} - \text{true_count})$	$MAE < 5$ per trap image
Species classification	Top-1 accuracy on IP102	$>60\%$ zero-shot (vs 80% supervised)
Tiny object detection	AP at IoU 0.25 (tiny objects)	>0.4 AP for $<32\text{px}$ objects
Alert timeliness	Time from image upload to alert	<5 seconds on T4
Threshold accuracy	% of correct above/below threshold decisions	$>90\%$ accuracy

Table 3.2: Evaluation metrics for Sticky Trap Auditor.

3.5 Feasibility & Risks

GPU: Colab T4 (same). **Data:** IP102 freely available (GitHub + Kaggle). **Key Risk (HIGH):** Tiny object detection is the hardest challenge across all alternatives. Grounding DINO is optimized for medium/large objects; its performance on $<32\text{px}$ insects may be very poor. **Mitigation:** (1) Multi-scale inference: run Grounding DINO at 1x, 2x, 4x zoom and merge detections. (2) Tiling: split the sticky trap image into overlapping tiles. (3) If zero-shot detection fails below 0.3 AP, fine-tune Grounding DINO on 50 annotated sticky trap images (still minimal labeling). **Timeline:** 14 weeks. **Cost:** \$0. **This is the highest-risk alternative** but also the most publishable if the refinement loop succeeds on tiny objects.

3.6 Publishability

Primary Venue: Computers and Electronics in Agriculture (Elsevier, IF ~8.3) or ESA (Entomological Society of America) journals. **Novelty Hook:** Proving zero-shot tiny-object detection and counting on sticky traps with multi-agent refinement is a massive contribution. The IP102 benchmark (798 citations) gives instant credibility. The practical impact (automated pest monitoring replacing manual counting) is valued by agricultural engineering journals.

3.7 Condensed Roadmap

Weeks	Tasks	Lead
1-2	Setup + IP102 download + test Grounding DINO on sticky trap images	All
3-5	Multi-scale inference pipeline + SAM 2 per-insect segmentation	M1+M2
6-8	Counting logic + ChromaDB pest thresholds + alert generation	M1+M3
9-11	Evaluation on IP102; tiny-object ablation; comparison to Li 2021	M3
12-14	Demo (upload trap photo, view count + alert) + thesis	M4+All

Table 3.3: Condensed 14-week roadmap.

4. Alternative: Post-Harvest Grader (Conveyor Belt Sorting)

4.1 Problem & Motivation

After harvest, produce (potatoes, apples, mangoes) must be sorted by quality. Bruised or rotten items contaminate the batch and reduce shelf life. Manual sorting is slow and inconsistent across shifts. Post-harvest loss reduction directly addresses UN Sustainable Development Goal 12.3: halve per-capita food waste by 2030. The Surface Defect Apple (SDA) dataset (Lee et al., 2023, cited by 30) contains 12,000 images with 6,201 normal and 3,799 defect images. Ryu et al. (2026, cited by 3) applied deep learning semantic segmentation to Fuji apples for four defect types (cracks, bruises, diseases, scars). A GitHub repository (graccy4eva/anthracnose_disease) contains potato tuber defect data: 118 germinations, 219 common scabs, 59 bug-eyes, 83 dry rots, and 80 bruises across 3,000+ images. Liang et al. (2022, cited by 80) achieved real-time apple defect grading using BiSeNet V2.

4.2 Pipeline

Input: Conveyor belt image of produce + prompt. **Grounding Agent:** "bruise. rot. fungal spot. healthy potato. defect." produces bounding boxes with defect labels. **Segmentation Agent:** SAM 2 segments each defect precisely for severity grading (defect area as % of total surface area). **Analysis Agent:** Acts as a virtual QA manager: computes "Grade A" (defect area < 5%) vs "Grade B" (5-15%) vs "Grade C" (>15%, reject). Estimates shelf-life reduction per defect type. **Report Agent:** Outputs daily yield report (Grade A/B/C counts) + per-item defect masks + shelf-life prediction.

4.3 Datasets

Dataset	Content	URL	Role
Surface Defect Apple (SDA)	12,000 images, 6201 normal + 3799 defect	PMC (Lee 2023, cited 30)	Primary
Potato Tuber Defects	118 germ + 219 scab + 59 bug-eye + 83 rot + 80 bruise	github.com/graccy4eva	Primary
Apple Sorting (Roboflow)	Object detection format	universe.roboflow.com	Supplementary
Kaggle Potato Disease	2152 leaf images (early/late blight)	kaggle.com	Supplementary

Table 4.1: Datasets for Post-Harvest Grader.

4.4 Evaluation Metrics & Targets

Metric	Computation	Target
Defect detection	AP at IoU 0.5 for each defect type	>0.6 AP zero-shot
Severity grading	Accuracy of Grade A/B/C classification	>80% accuracy

Metric	Computation	Target
Shelf-life prediction	Correlation with actual shelf-life	$r > 0.7$
Sorting throughput	Images processed per minute	>30 images/min on T4
Cross-produce transfer	Same pipeline on potato AND apple	>70% accuracy on both without retraining

Table 4.2: Evaluation metrics for Post-Harvest Grader.

4.5 Feasibility & Risks

GPU: Colab T4 (same). **Data:** SDA dataset (PMC, freely available). Potato defects on GitHub. **Key Risk (MEDIUM):** Defect appearance varies significantly across produce types (potato scabs look nothing like apple bruises). Grounding DINO's zero-shot capability may handle this well since it understands "bruise" and "rot" as general concepts. **Mitigation:** Start with one produce type (apple, which has the most data) and demonstrate cross-produce transfer as the novelty. **Timeline:** 14 weeks. **Cost:** \$0.

4.6 Publishability

Primary Venue: MDPI Sustainability (IF ~3.9) or Postharvest Biology and Technology (Elsevier, IF ~7.0). **Novelty Hook:** SDG 12.3 alignment is a major advantage; papers with SDG impact receive favorable review. The zero-shot cross-produce transfer (same pipeline grades both potatoes and apples without retraining) is the key novelty. Existing sorting papers are all single-produce, single-defect-type supervised classifiers.

4.7 Condensed Roadmap

Weeks	Tasks	Lead
1-2	Setup + SDA + potato defect download; test GDINO on produce images	All
3-5	Pipeline: "bruise. rot. defect. healthy." prompts + SAM 2 masks	M1+M2
6-8	Severity grading (Grade A/B/C) + shelf-life estimation	M1+M3
9-11	Evaluation on SDA + potato; cross-produce transfer test	M3
12-14	Demo (upload conveyor image, view grade report) + thesis	M4+All

Table 4.3: Condensed 14-week roadmap.

5. Alternative: Hydroponics Root Monitor (Indoor Farming)

5.1 Problem & Motivation

In vertical farms and hydroponic systems, roots are submerged in opaque nutrient solution and cannot be visually inspected. Root rot (brown necrotic roots from *Pythium* or *Fusarium*) develops silently and kills the crop before above-ground symptoms appear, causing devastating losses in high-density indoor operations. Controlled Environment Agriculture (CEA) is booming globally with multi-billion dollar investment (Rathor et al., 2024, cited by 149; Dohlman et al., 2024, cited by 34). Computer vision in CEA enables automatic crop monitoring (Kofi et al., 2026). Ultralytics published a blog on vision AI for hydroponic farming (2025). Bao et al. (2025, cited by 16) demonstrated zero-shot instance segmentation for plant phenotyping in vertical farms. DeepGreen (Rathor et al., 2026, cited by 3) is a real-time deep learning system for smart vertical farming disease identification. Proving vision AI works underwater/in hydroponic conditions is novel and directly relevant to the CEA industry.

5.2 Pipeline

Input: Image of root zone (through transparent grow tube or pulled from solution) + prompt. **Grounding Agent:** "white healthy root. brown necrotic root. root rot. healthy root mass." produces bounding boxes. **Segmentation Agent:** SAM 2 segments each root strand despite refraction, turbidity, and water distortion. **Analysis Agent:** Computes root health ratio (brown_area / total_root_area). Correlates with mock IoT sensor data (pH, dissolved oxygen, EC, water temperature) from a mock database. Predicts disease 3 days before visible above-ground symptoms based on root color trend. When root rot detected, queries SmartDesk's KnowledgeAgent for treatment protocols. **Report Agent:** Outputs root health heatmap + sensor correlation + treatment recommendation.

5.3 Datasets

Custom Dataset (Required): No public hydroponics root rot dataset exists at scale. The team must collect approximately 50-100 photos: (1) 25-50 healthy white root images from a local hydroponics facility or YouTube videos with permission, (2) 25-50 brown/necrotic root images from the same sources. This is feasible for zero-shot: SAM 2 proves it works without training data on this novel domain. The custom dataset serves purely as evaluation ground truth, not training data. **Supplementary:** PlantVillage (leaf-level disease images for above-ground symptom correlation) and any vertical farming imagery from public sources (Omdena blog on AI in vertical farming, 2026).

5.4 Key Papers

Rathor et al. (2024, cited by 149): Empowering vertical farming through IoT and AI-driven disease detection. **Bao et al. (2025, cited by 16):** Zero-shot instance segmentation for plant phenotyping in vertical farms using Grounding DINO + SAM. **Kofi et al. (2026):** AI in CEA for automatic crop monitoring and disease detection. **DeepGreen (Rathor et al., 2026, cited by 3):** Real-time DL system for vertical farming disease identification. **Jones et al. (2026):** Deep learning + computer vision for lettuce growth monitoring in vertical farms. **Nawshad et al. (2025, cited by 3):** Robust hydroponic system with automatic disease detection.

5.5 Evaluation Metrics & Targets

Metric	Computation	Target
Root segmentation	IoU (healthy vs necrotic root mask)	>0.6 IoU zero-shot
Health ratio accuracy	Correlation with expert root health rating	r > 0.8
Early prediction	Days before above-ground symptoms	>2 days advance warning
Sensor correlation	Correlation between root health and pH/DO/EC	r > 0.6 for at least one sensor
Treatment retrieval	RAG relevance of recommended treatment	>80% relevant recommendations

Table 5.1: Evaluation metrics for Hydroponics Root Monitor.

5.6 Feasibility & Risks

GPU: Colab T4 (same). **Data (MEDIUM RISK):** Custom data collection adds 1-2 weeks. No public dataset means evaluation is limited to 50-100 images, which may concern reviewers. **Key Risk (HIGH):** (1) Root imagery through water has severe refraction/turbidity, degrading both Grounding DINO and SAM 2 performance. (2) Limited evaluation data may weaken statistical significance. **Mitigation:** (1) Test on both in-water and out-of-water root images; the out-of-water case (roots pulled from solution momentarily) is easier and more practical. (2) Supplement with leaf-level disease images from PlantVillage for the above-ground symptom correlation. (3) Partner with a local hydroponics facility for data collection. **Timeline:** 14 weeks (including 2 weeks for data collection). **Cost:** \$0.

5.7 Publishability

Primary Venue: MDPI AgriEngineering or Smart Agricultural Technology (Elsevier). **Novelty Hook:** "Zero-shot root rot detection in hydroponics" is genuinely novel. No existing paper combines foundation model segmentation with

root-level monitoring in CEA. The sensor correlation (vision + IoT) and early prediction angle (3 days before above-ground symptoms) are strong practical contributions. CEA is a multi-billion dollar industry hungry for automated monitoring solutions. **Weakness:** Small custom dataset (50-100 images) limits statistical power. Address this honestly in the paper and frame the contribution as a proof-of-concept validated on the available data.

5.8 Condensed Roadmap

Weeks	Tasks	Lead
1-2	Setup + collect 50-100 root images (healthy + necrotic) from hydroponics facility	All
3-5	Pipeline: "healthy root. root rot." prompts + SAM 2 masks on root images	M1+M2
6-8	Root health ratio + mock IoT sensor correlation + KnowledgeAgent RAG	M1+M3
9-11	Evaluation on custom dataset; early prediction validation; RAG quality	M3
12-14	Demo (upload root image, view health report + treatment) + thesis	M4+All

Table 5.2: Condensed 14-week roadmap.

6. Comparative Summary & Decision Framework

The following table provides a side-by-side comparison of all 5 alternatives and the main 5-app AgriAgent project, enabling the team and Prof. Dhore to make an informed decision. Each alternative uses the same technology stack (Grounding DINO + SAM 2 + LangGraph) but focuses on a single application with deeper contribution.

Alternative	Focus	Datasets	Risk	Venue	Time	Publishability Hook
Main: 5-App AgriAgent	5 apps, breadth	SugarBeets, DeepWeeds, PlantDoc, GWHD, PlantVillage	MEDIUM	CVPR Agri-Vision	16 weeks	Broad zero-shot generalization
1: Drone Scout	Orthomosaic weed mapping	WeedMap, SugarBeets	LOW	Remote Sensing / C&EA;	14 weeks	Macro-level segmentation trend
2: Smart Harvester	Fruit ripeness grading	MinneApple, Strawberry IEEE	LOW	IEEE RA-L / CASE	14 weeks	Robotics grasp + NL control
3: Sticky Trap	Insect pest counting	IP102, Sticky trap images	HIGH	C&EA; / ESA	14 weeks	Tiny object detection flex
4: Post-Harvest	Conveyor belt sorting	SDA Apple, Potato defects	MEDIUM	Sustainability / PBT	14 weeks	UN SDG 12.3 food waste
5: Hydroponics	Root rot monitoring	Custom (50-100 images)	HIGH	AgriEngineering / SAT	14 weeks	CEA industry novelty

Table 6.1: Comparative summary of all alternatives.

Recommendation: If Prof. Dhore approves the main 5-app AgriAgent, proceed with that (broadest contribution). If he requests a narrower focus, Alternative 4 (Post-Harvest Grader) offers the best risk-to-novelty ratio: medium risk, strong SDG alignment, and clear cross-produce zero-shot transfer novelty. Alternative 1 (Drone Scout) is the safest fallback with excellent data availability. Avoid Alternative 5 (Hydroponics) unless the team has confirmed access to a hydroponics facility for data collection.